Understanding Cross-Domain Generalization in Semantic Segmentation: A Study Using Aerial Rooftop Imagery



GROUP 10

Milestone 2: Dataset Preparation

Course: DSAI Labs

Team: Team 10

Term: September 2026

Change Log

This log is maintained for every submitted document, as required in the Milestone 1 review. It records what was decided or corrected and confirms that the change is incorporated in this document.

#	Item	What was done	Where	Status
1	Milestone 2 report prepared	This report follows the 13-section structure given in the course guidelines for Milestone 2 (Dataset Preparation): identification, description, governance, exploratory analysis, preprocessing, integration, augmentation, splitting, the final prepared dataset, challenges, deliverables, and next steps.	Sections 1-13	Incorporated
2	Dataset decisions carried forward from Milestone 1	The core datasets are WHU (aerial), Inria and Massachusetts, giving a 3x3 train-test matrix. AIRS is used only as a small test-only subset taken from its validation split, because its official test labels are not public. AIRS is never used for training.	Sections 2, 9	Incorporated
3	Correction: AIRS split sizes	The Milestone 1 draft listed the AIRS test split as 95 tiles. The original paper (Chen et al., 2019) gives 857 train / 94 validation / 96 test, which sums to the stated total of 1,047 tiles. The figure is corrected to 96 in this report.	Sections 2, 3	Incorporated
4	Clarification: Massachusetts annotations	The annotations are building footprints. The target maps were made by rasterising building footprints obtained from OpenStreetMap, to which the City of Boston contributed footprints for the whole city (Mnih, 2013). Massachusetts is not a roof-outline dataset.	Sections 3, 4	Incorporated
5	Note: WHU tile total	The official WHU page states 8,189 tiles in its text, while its own split counts (4,736 + 1,036 + 2,416) sum to 8,188, which is also the total in the Zenodo deposit of the dataset. Both figures are reported in Section 3 instead of choosing one silently.	Section 3	Incorporated
6	Measured preparation results incorporated	After the preparation run on the team server, this report was updated with the measured results: the per-split class distribution and empty-tile shares and the EDA charts (Section 5), the lossless mask-conversion proof (Section 6), the confirmed final counts of 23,372 tiles with all readiness checks passed (Section 10), and the real problems met during the run (Section 11).	Sections 5, 6, 10, 11	Incorporated
7	AIRS deferred	AIRS was not downloaded in this milestone (team decision). Statements that described an AIRS subset as already prepared have been corrected throughout: AIRS remains a planned test-only subset from its validation split, to be prepared in a later milestone.	Sections 2, 5-7, 9-13	Incorporated

1. Introduction

Milestone 1 defined the project and reviewed the literature. The project studies why a building and rooftop segmentation model that scores well on its own dataset loses accuracy when it is tested on a different dataset. Its main research question, adopted in the revised Milestone 1, is: to what extent can differences in image resolution, appearance, and annotation boundaries help explain cross-dataset performance degradation in rooftop segmentation, and what limitations remain in attributing the degradation to individual factors? The committed goal (Objective 1) is to measure the degradation under one fixed protocol and to investigate the sensitivity of the model to selected factors. Cleanly isolating and quantifying each factor (Objective 2) is not promised.

The objective of Milestone 2 is data. This milestone identifies the datasets, downloads them to the team server, describes and checks them, and prepares them in one common, training-ready format, so that model work in Milestone 3 can start without any further data preparation.

2. Dataset Identification

Four datasets are used. Three are core datasets (trained on and tested on). The fourth, AIRS, is planned only as a small test-only subset, for the one contrast the core datasets do not provide: roof-outline labels. AIRS was not downloaded in this milestone; its subset will be prepared in a later milestone (Sections 9-11).

Database Name, Source, Status and Purpose:

Dataset	Source and download	Status	Purpose in this project
WHU Building Dataset (aerial subset)	Photogrammetry and Computer Vision Group (GPCV), Wuhan University. Dataset Link	Public, direct download	Big set of aerial photos with Hand Checked building masks.
Inria Aerial Image Labeling	Inria project site. Dataset Link	Public, form-based download	Covers cities in the USA and Austria; the 5 labelled training cities are used.
Massachusetts Buildings	Volodymyr Mnih, University of Toronto. Dataset Link	Public, direct download	Covers cities and suburbs of the Boston area with mixed images.
AIRS	Chen et al. (2019), ISPRS Journal. Dataset Link	Not downloaded in this milestone - deferred	Planned test-only contrast dataset (roof-outline labels) for a later milestone. Never used for training.

Why each dataset was selected:

WHU (aerial):

- We selected the WHU aerial building dataset because it has the cleanest labels. Its building masks were edited by hand, so the mask edges follow the buildings closely.
- Its pixel size is 0.3 m, the same as Inria. So when we compare WHU and Inria the resolution stays the same. Because all images come from one city (Christchurch, New Zealand), the model gets a strong score on WHU.

Inria:

- We selected the Inria Aerial Image Labeling dataset because it covers many cities in the USA and Austria, so it has more variety in building style.
- Its pixel size is 0.3 meter, the same as WHU. Their masks come from official building footprints, an official-footprint label style. This helps us study annotation boundary differences across the datasets.
- Test masks are hidden so we make our own validation split from the training images.

Massachusetts:

- The Massachusetts dataset is a public and widely used benchmark with clear aerial images. Its masks come from OpenStreetMap, so they cover a large area.
- The images cover the Boston area, with dense city and suburbs so the building density changes across the data.

Alternatives considered:

- **SpaceNet, DeepGlobe and AIRS:** Satellite based dataset with different labels. So they add more differences that we cannot control.
- **Microsoft Building Footprints and Google Open Buildings:** They give building outlines only without matching images and masks that we can train on directly.
- **WHU-Mix:** Larger and more varied, but it is built partly from WHU and Inria data, so it overlaps with datasets we already use.

3. Dataset Description

The task is binary semantic segmentation. Each sample is a pair of an RGB aerial image and a mask of the same size in which each pixel is building/rooftop (1) or background (0).

	WHU (aerial)	**Inria**	**Massachusetts**
Image Size	512 × 512 pixels	5000 × 5000 pixels	1500 × 1500 pixels
Total Images	8,188 images	360 images (180 with public labels, used here)	151 images
Image Type	TIFF	TIFF	TIFF
Pixel Size	0.3 meter	0.3 meter	1 meter

Image Label	Mask (with test) (black and white)	Mask (without test)	Mask (with test) (black and white)

Real size of Dataset:

	WHU	**Inria**	**Massachusetts**
Pixels per side	512	5000	1500
Pixel size	0.0003 km (0.3 m)	0.0003 km (0.3 m)	0.001 km (1 m)
Width	0.1536 km	1.5 km	1.5 km
Height	0.1536 km	1.5 km	1.5 km
Area of one image	0.0236 km²	2.25 km²	2.25 km²
Number of images	8188 images	360 images	151 images
Total area	193.2 km²	810 km²	339.75 km²
Comparable to Indian Cities	A little smaller than Kolkata (about 206 km²)	Larger than Bengaluru (about 712 km²)	A little smaller than Lucknow (about 348.8 km²)

Number of Features:

The three datasets contain images not tables so there are no feature columns. The pixels of the image are the input features

- **Input:** 1 aerial image with 3 channels, red, green and blue (RGB). Each pixel value is from 0 to 255. The image is stored as a TIFF (.tif) file.
- **Target:** The mask is stored as a TIFF (.tif) file.

Target Variable:

The target variable is a binary mask with the same height and width as the input image. Each pixel has one of two values(after scaling):

- 1 = building (rooftop in white color).
- 0 = background (road, tree, grass in black color).

Sample records:

Each Datasets contains image and mask files with the same name in different folders.

Dataset	**Image file**	**Mask file**
WHU	image/1.tif	label/1.tif
Inria	images/austin1.tif	gt/austin1.tif
Massachusetts	train/sat/22678915_15.tiff	train/map/22678915_15.tif

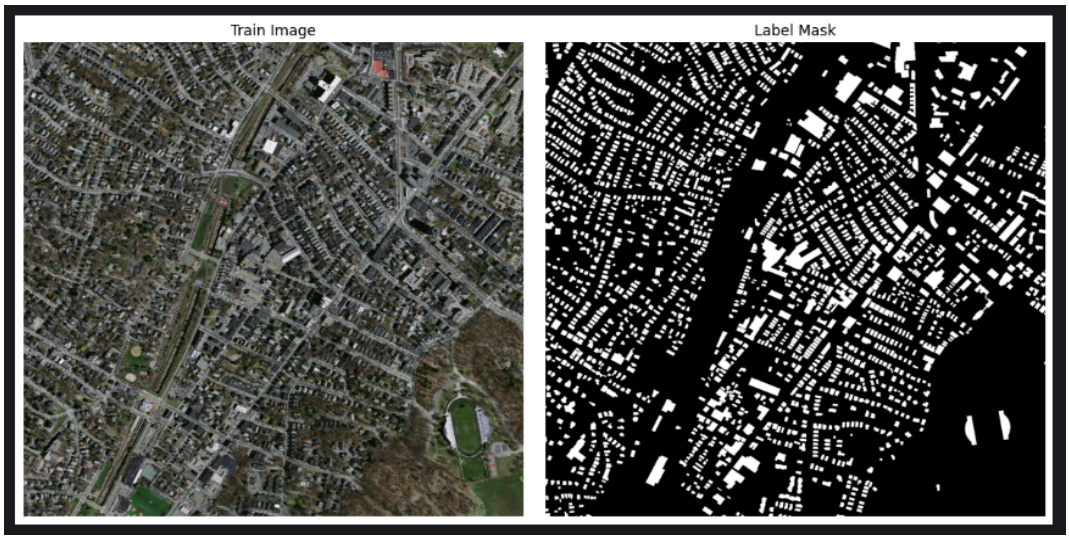


Figure 1.1 sample record

The left picture is the colour aerial photo in (RGB). The right picture is its mask. White pixels are buildings and Black pixels are background.

Images Descriptions and Format:

Item	Description
Input image	Colour aerial photo and 3 channels(RGB).
Ground truth mask	One channel, black and white.
Location	WHU: Christchurch, New Zealand. Inria: 10 cities in the USA and Austria. Massachusetts: Boston area, USA.
Label source	WHU: Hand Edited Footprints. Inria: Official Footprints. Massachusetts: OpenStreetMap Footprints.

Dataset schema:

Dataset	Image folders	Mask folders
WHU	train/image, val/image, test/image	train/label, val/label, test/label
Inria	train/images, test/images	train/gt (no test masks)
Massachusetts	train/sat, valid/sat, test/sat	train/map, valid/map, test/map

4. Data Governance

Data source and licensing

Dataset	Source	Who owns or released it	Permission to use
WHU	Aerial imagery of Christchurch, New Zealand	Released by Wuhan University (Ji et al., 2019)	Public, released for research.
Inria	Aerial imagery, Building footprints of USA and Austria	Released by Inria, France (Maggiori et al., 2017)	Public, released for research..
Massachusetts	Aerial imagery of the Boston area.	Released by the University of Toronto (Mnih, 2013)	Public, released for research.

Privacy

- **Is there personal data (PII):** No names, faces, phone numbers or other identifiers. The data is only aerial images and building masks.
- **Can people be identified:** No. At 0.3 m to 1 m per pixel, a person is not visible and a car is only a few pixels.

Ethics and bias:

- **Geographic bias:** The data covers New Zealand, the USA and Austria. None of it is from India.
- **Small buildings:** Small buildings and dense areas are more likely to be missed, especially at 1 meter.

Reproducibility and Compliance:

- **Dataset Versions:** For each dataset (WHU, Inria and Massachusetts), we record the source link we downloaded.
- **Splits:** We use the official train, validation and test splits for WHU and Massachusetts. Inria's test labels are not public, so we make our own split from its 180 labelled tiles.
- **Preprocessing:** This covers tiling images to a common size and converting masks to 0 and 1, applied identically to all three datasets. Image appearance is deliberately not normalised or corrected, because appearance differences are one of the factors the project studies.
- **Evaluation:** We use the same evaluation code and the same metrics (IoU, precision, recall and F1) for every experiment.
- **Licence and copyright:** These are public datasets. We use the datasets only for this academic project.

5. Exploratory Data Analysis (EDA)

For image data, EDA means examining the images and masks themselves: how much of each dataset is building, how the buildings are distributed, what the label quality is known to be, and where the datasets differ. The findings below come from the official documentation of each dataset, from the checks run during preparation (Section 6), and from the metadata recorded for every prepared tile.

Dataset summary

- Scale: the core datasets together provide 8,188 WHU tiles, 180 labelled Inria images and 151 Massachusetts images before tiling. After preparation (Section 10) the project works with 23,372 prepared tiles in total across the three core datasets. AIRS was not downloaded in this milestone, so the prepared dataset contains no AIRS tiles; the AIRS test-only subset is deferred to a later milestone.
- Resolution spread: 0.3 m (WHU, Inria), 1 m (Massachusetts) and 0.075 m native for the AIRS source imagery, which is not part of this milestone; when the AIRS subset is prepared later it will be resampled to 0.3 m for the contrast test. This spread is deliberate: it is one of the factors of the study.
- Building counts (documented): WHU records 130,500 buildings in its training tiles, 14,500 in validation and 42,000 in test. AIRS records 226,342 buildings after label refinement. The other datasets do not publish a single building total.

Class distribution

The target has two classes, building and background, and in all four datasets background dominates: buildings are a minority of pixels in typical tiles, and some tiles (open land, parks, water edges) contain no building at all. The prepared metadata records the building-pixel count of every tile, which gives the exact class distribution of the prepared dataset per split, and flags empty tiles. Two consequences follow for later milestones: pixel accuracy is not a useful measure on this data (Milestone 1 already states this), and empty tiles are kept in the test sets, because removing them would inflate test scores, while their share in training lists is controlled through the metadata.

Measured class distribution of the prepared tiles (computed from metadata.csv during preparation):

Dataset and split	Tiles	Building pixels	Building share	Empty tiles (no building)
WHU train	4,736	231,714,558	18.66%	419 (8.85%)
WHU validation	1,036	29,873,959	11.00%	299 (28.86%)
WHU test	2,416	70,478,342	11.13%	685 (28.35%)
Inria train	10,530	452,023,437	16.38%	1,578 (14.99%)
Inria validation	2,025	80,417,534	15.15%	461 (22.77%)
Inria test	2,025	72,934,646	13.74%	447 (22.07%)

Massachusetts train	548	19,020,177	13.24%	21 (3.83%)
Massachusetts validation	16	505,021	12.04%	0 (0%)
Massachusetts test	40	1,753,417	16.72%	0 (0%)

No tile in any split is fully covered by buildings. The measured EDA results are in the table above and the charts below.

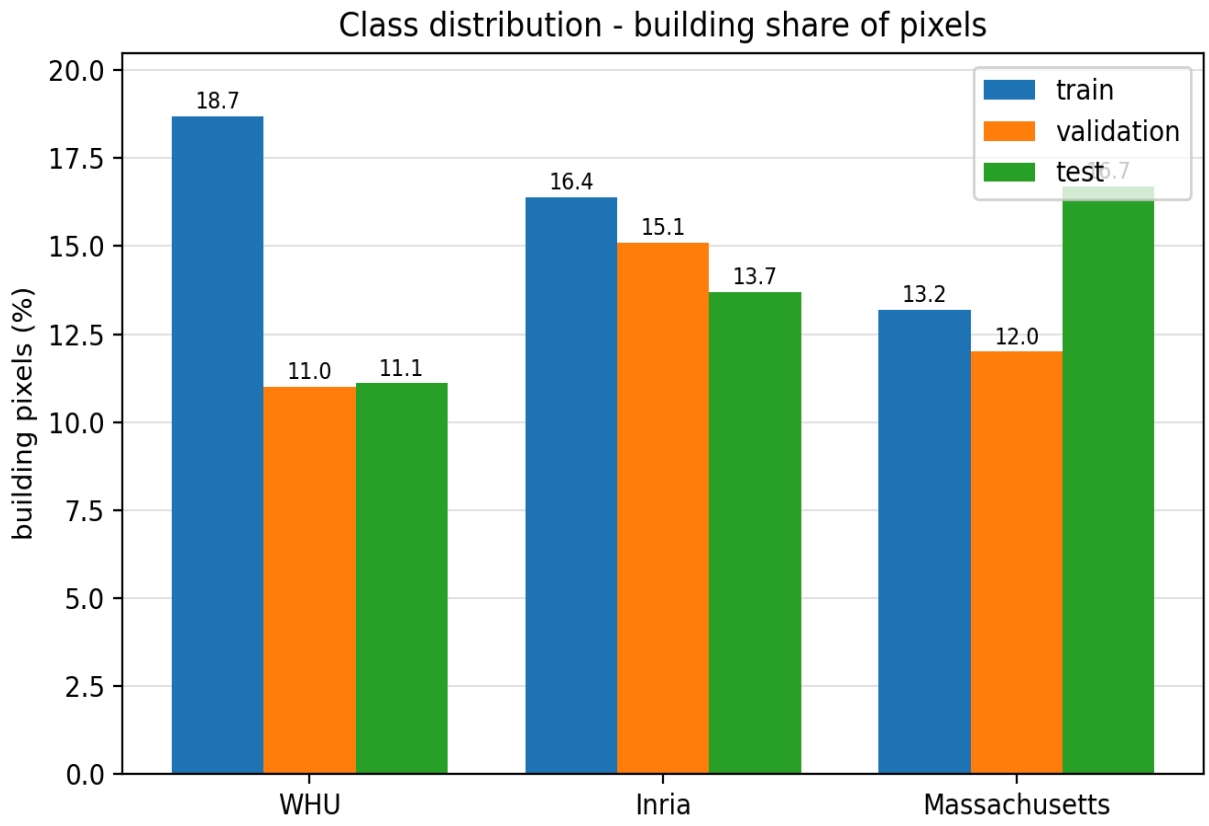


Figure: Building share of pixels in each dataset and split.

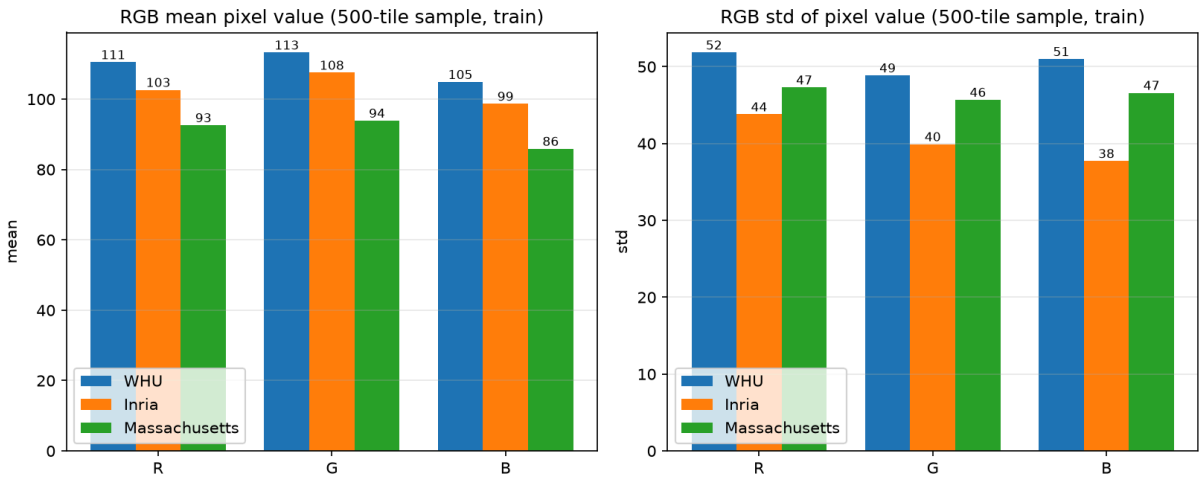


Figure: Mean and spread of RGB pixel values (train splits, 500-tile sample).

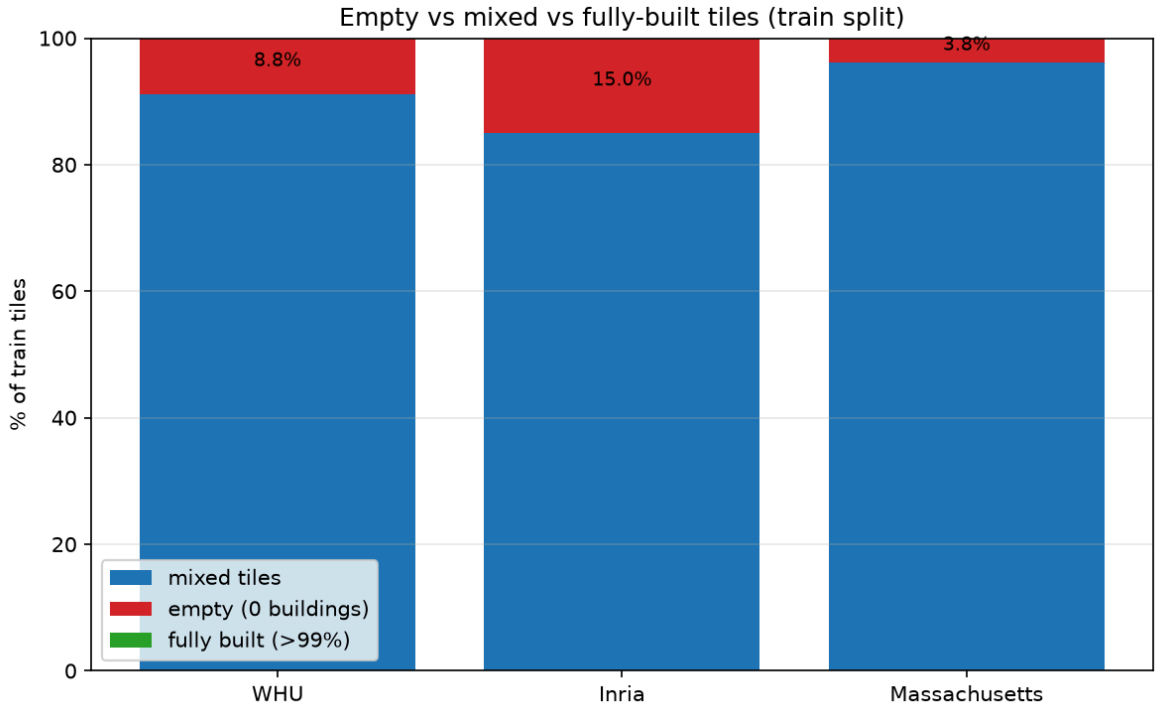


Figure: Empty, mixed and fully-built tiles in the train splits.

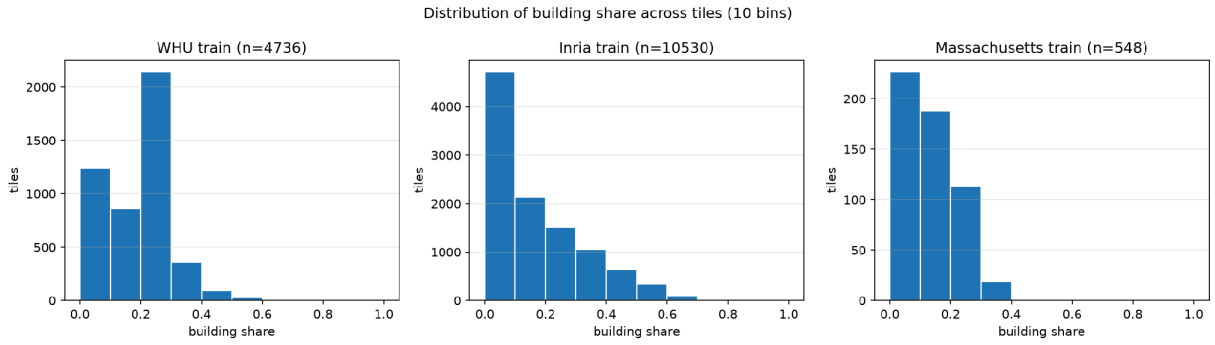


Figure: Distribution of building share across tiles (train splits).

Label quality and missing or inconsistent data

- Massachusetts: the thesis reports that the data was restricted to regions with an average omission noise of roughly 5% or less (some buildings present in the image are missing from the labels), and that the test and validation target maps were hand-corrected. Training labels therefore contain a small, documented amount of missing-building noise.
- Inria: the official test labels are not public at all, which is a missing-data problem at the dataset level. It is handled by the documented split of Section 9, not by estimating the missing labels.
- AIRS: the official test labels are withheld, and AIRS was not downloaded in this milestone. When the subset is prepared later, only validation-split tiles, whose labels are public, will be used.
- WHU: labels were manually edited by the dataset team, and this is the cleanest label source of the four; the one inconsistency found is in the documentation itself (the tile total, Section 3), not in the tiles.
- Pairing and format inconsistencies found during preparation: mask encodings differ by dataset (Inria masks use 255 for building, WHU and Massachusetts use their own target-map encodings), and file formats differ (GeoTIFF against PNG-style tiles). These are standardised in preprocessing (Section 6). True duplicate images were not found within any dataset. The final pairing check on the prepared data confirms this: all 23,372 prepared tiles are paired, with no orphan images, no orphan masks, no corrupt files and no size mismatches in the final dataset.

Duplicates, overlap and outliers

One overlap will exist between datasets when the AIRS subset is prepared in a later milestone: WHU and AIRS both cover Christchurch, and their scenes may partly overlap. The AIRS subset will therefore be kept strictly separate as a test-only set, never merged into any training data, and its results will carry this caveat (as stated in Milestone 1). Within datasets, "outlier" tiles are of three kinds: tiles with no buildings, tiles that are almost fully covered by one large building or roof complex, and tiles with heavy shadow or tree cover over roofs. These are not removed -

they are exactly the hard cases the error analysis in later milestones needs - but they are identifiable through the metadata (building-pixel counts and source positions).

Visual examination

Sample image-mask pairs from every dataset and split were overlaid and inspected during preparation (5 random tiles per dataset and split, 45 overlays in total). The masks sit correctly on the buildings in all three datasets, and no flipped, shifted or wrongly paired example was found. The documented differences are visible, not just numerical: Massachusetts tiles look visibly coarser at 1 m and show speckle inside buildings consistent with its known label noise of about 5%; Inria cities differ in building density and roof colour between the US and Austrian cities. The overlay images and the scripts that produced these checks are listed in Section 12.

6. Data Preprocessing

The benchmark datasets are already cleaned and labelled by their authors, so no heavy cleaning is needed. The preprocessing in this project is the smaller, exact set of steps needed to turn four differently packaged downloads into one uniform, training-ready dataset. The steps are:

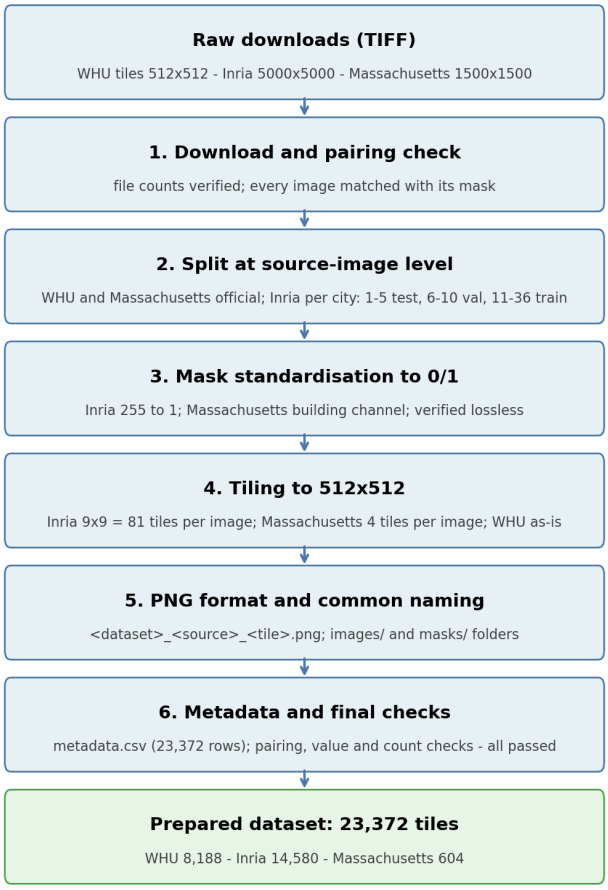


Figure: The preparation pipeline applied to all three datasets, in order.

- Download and integrity check: files are downloaded from the official sources of Section 2 to the team server, unzipped, and checked - file counts against the documented counts, and every image checked to open correctly.
- Image-mask pairing: each image is matched with its mask. Pairs with a missing partner or a size mismatch are set aside and recorded; none are silently dropped.
- Tiling to 512x512: WHU tiles are already 512x512 and are used as they are. Inria images (5000x5000) are cut into a 9x9 grid of non-overlapping 512x512 tiles (using the top-left 4608x4608 region; the remaining border strip of each image is not used, and this rule is recorded). Massachusetts images (1500x1500) are cut into 4 non-overlapping 512x512 tiles each, with the remaining border strip not used. Tiling is done after splitting, at source-image level, so tiles from one source image can never fall into two different splits.
- AIRS subset preparation: not performed in this milestone. AIRS was not downloaded, by the team's decision, and no AIRS folder or metadata rows were created. The planned subset (10-20 source tiles from the validation split only, each resampled from 0.075 m to 0.3 m and then tiled to 512x512 in the same way) is deferred to a later milestone.
- Mask standardisation: every mask is converted to a single channel with values 0 (background) and 1 (building) only - for example Inria masks are converted from 255 to 1. This is an encoding change only; no label is redrawn, moved or corrected. The conversion was verified to be lossless: the number of building pixels was counted before and after conversion and was identical for every dataset (WHU: 231,714,558 building pixels in train, 29,873,959 in validation and 70,478,342 in test; Inria: 710,215,250 over the source images; Massachusetts: 45,841,465 over the source images).
- Format standardisation: all tiles are saved as PNG, images as 3-channel RGB and masks as single channel, with the common naming and folder structure of Section 7.
- Pixel scaling: image pixel values are scaled to the range 0-1 when the tiles are loaded for training. Nothing else is done to image appearance - no colour correction, no histogram equalisation, no sharpening. This is deliberate: appearance differences between datasets are one of the factors the project studies, and "cleaning" them away in preprocessing would destroy part of the evidence.
- Metadata generation: for every prepared tile, the metadata row of Section 3 is written, including its building-pixel count and an empty-tile flag. The final metadata.csv contains 23,372 rows, exactly one per prepared tile.
- Visual verification: a sample of prepared pairs from each dataset and split is overlaid and checked, as described in Section 5.

7. Dataset Integration

The three core datasets and the AIRS subset are integrated into one prepared dataset with a single structure. Integration here means one format and one index - the datasets are not mixed into a single pool, because which dataset a tile comes from is itself information the experiments need.

Prepared dataset	Folder layout
WHU/ Inria / Mass	train / validation / test, each with images/ and masks/ folders of PNG tiles
metadata.csv	One row per tile, with the schema of Section 3

Schema alignment

- Tile size aligned: 512x512 everywhere. Channels aligned: RGB images, single-channel masks. Mask encoding aligned: 0/1 everywhere. Naming aligned: <dataset>_<source>_<index>.png for image and mask.
- Resolution deliberately not aligned: Tiles keep their native ground resolution (0.3 m or 1 m), because resolution is a study factor. The AIRS subset, when it is prepared in a later milestone, will be the single exception: it will be resampled to 0.3 m by design, so that its contrast with WHU is about label convention rather than resolution (with the remaining confounds documented in Milestone 1).
- Annotation definitions not merged: footprint-style and roof-outline labels keep their identity through the dataset field. No common relabelling is attempted.

Conflicts and deduplication

The conflicting attributes between the sources are the different mask encodings and file formats, resolved by the standardisation steps of Section 6, and the different split conventions, resolved by documenting each dataset's split in Section 9 rather than forcing one rule on all four. No duplicate tiles arise from integration: the core datasets cover different cities on different continents (Boston; US and Austrian cities; Christchurch), and the one future geographic overlap, AIRS with WHU, will be handled by keeping AIRS as a separate test-only set that is never merged into training data.

8. Data Augmentation

No augmented samples are stored in the prepared dataset. Augmentation in this project is applied at training time, inside the training pipeline of later milestones, so the prepared dataset itself stays a faithful, countable copy of the source data.

- Planned training-time augmentation (fixed in the Milestone 1 protocol, the same for every run): horizontal and vertical flips and 90-degree rotations. Aerial images have no natural "up", so these change nothing about the content while multiplying the effective orientations.
- Not part of Milestone 2: the appearance and resolution changes used later as factor experiments (brightness and colour changes, resampling) are experimental interventions defined in Milestone 1. They are applied to test data at evaluation time in later milestones and are not baked into this prepared dataset.

9. Dataset Splitting

Dataset	Train	Validation	Test	Basis
WHU (aerial)	4,736 tiles	1,036 tiles	2,416 tiles	Official split, used as released
Inria	130 source images (tiles 11-36 of each of the 5 training cities)	25 source images (tiles 6-10 of each city)	25 source images (tiles 1-5 of each city)	Documented split
Massachusetts	137 images	4 images	10 images	Official split, used as released
AIRS (subset)	-	-	Deferred: not downloaded in this milestone; 10-20 source tiles from the validation split in a later milestone	Test-only; no train or validation set will be created from AIRS

The Inria split

Inria's official test labels are not public, so the official test set cannot be used for evaluation. Following the convention used in published work on this dataset (including Na et al., whose Inria evaluation also used labelled tiles from the training cities), the split is created inside the five labelled training cities: for each city, source tiles 1-5 form the test set, tiles 6-10 the validation set, and tiles 11-36 the training set. The dataset's own contest page likewise suggests holding out the first tiles of each training city for validation. The limitation is stated plainly, here and in Milestone 1: test tiles come from the same cities as training tiles, so Inria in-dataset scores will be optimistic compared with a truly unseen city.

Justification and leakage prevention

- Official splits are used wherever they exist (WHU, Massachusetts), so results stay comparable with published work.
- Splits are fixed before any tiling: the unit of splitting is the source image, and tiles are cut afterwards inside each split. No source image contributes tiles to two splits, which prevents the most common form of leakage in tiled image data.
- The AIRS subset cannot leak into training: when it is prepared later it will exist only as a test set, stored separately, and no model in the project will be trained on it. In this milestone no AIRS data was downloaded at all.
- No stratification by random sampling is used, because these benchmark splits are geographic or official by design; instead, the class record (building pixels per tile, Section 3) documents what each split actually contains.

10. Final Prepared Dataset

Applying the rules of Sections 6 and 9 gives the prepared dataset below. These are the final measured counts of the preparation run on the team server every image is paired with a same-named mask of matching size, every mask contains only the values 0 and 1, and the metadata row count (23,372) equals the tile count. All final readiness checks passed.

Dataset	Train tiles	Validation tiles	Test tiles	Total prepared tiles
WHU (aerial)	4,736	1,036	2,416	8,188
Inria	10,530	2,025	2,025	14,580
Massachusetts	548	16	40	604
AIRS (test-only subset)	-	-	-	Not prepared in this milestone (not downloaded)
Core total (WHU + Inria + Massachusetts)	15,814	3,077	4,481	23,372

- Final size: 23,372 prepared tiles for the three core datasets.

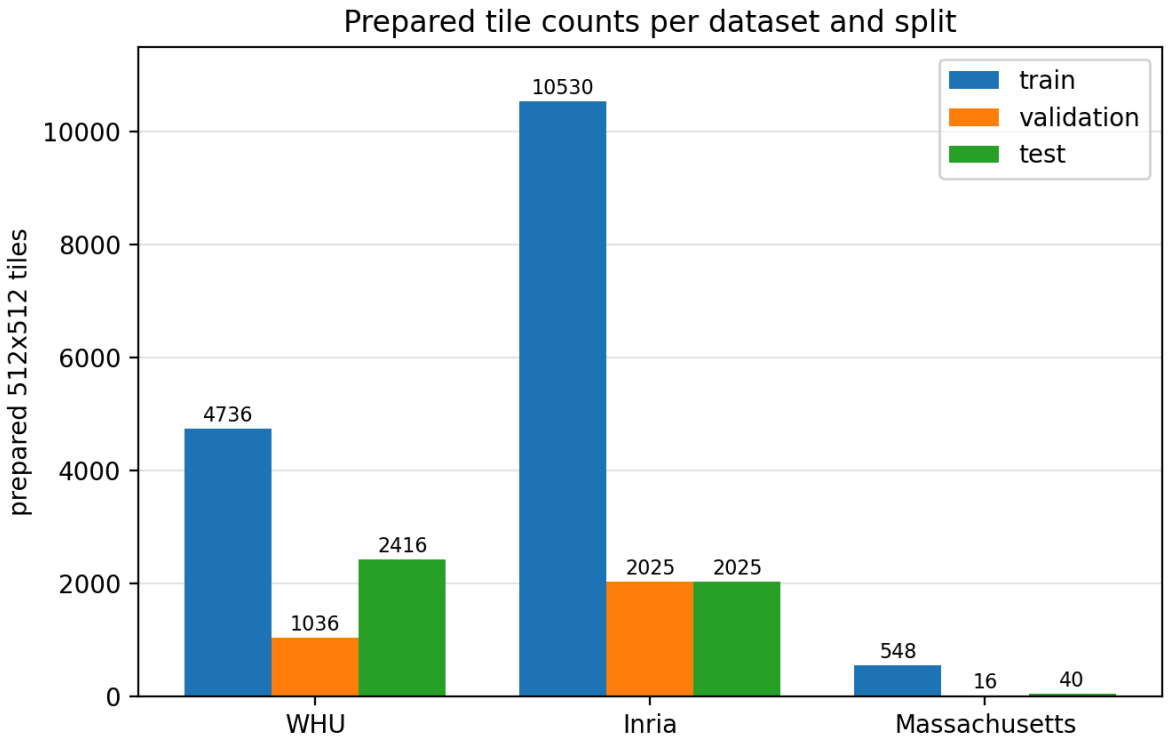


Figure: Prepared tile counts per dataset and split.

- Features: every sample is a 512x512 RGB image (3 channels) with a 512x512 binary mask. There are no additional features.

- Final class distribution: two classes, building (1) and background (0). Measured on the prepared tiles, buildings cover 18.66% of pixels in WHU train, 11.00% in WHU validation and 11.13% in WHU test; 16.38% in Inria train, 15.15% in Inria validation and 13.74% in Inria test; and 13.24% in Massachusetts train, 12.04% in Massachusetts validation and 16.72% in Massachusetts test. Background is therefore the majority class in every split. Empty tiles (no building pixel at all) range from 0% of tiles in Massachusetts validation and test to 28.86% in WHU validation, and no tile in any split is fully covered by buildings. The full per-split table is given in Section 5, and the tile-by-tile record is in metadata.csv.
- Summarising the preprocessing steps performed:
- Downloaded the files and checked their integrity
- Matched each image with its mask
- Standardised all three core datasets to 512x512 tiling
- Recorded AIRS as deferred: it was not downloaded in this milestone, and its test-only subset (from the validation split, resampled to 0.3 m) will be prepared in a later milestone,
- Standardised masks to 0s and 1s
- Standardised image formats to PNG
- Generated metadata
- Visually verified samples, as per the methodology described in section 5.

Following the aforementioned preprocessing steps, the prepared datasets are considered ready for model training. A training script would only need to access the folder structure, as well as the metadata.csv file, where each row names an image, its mask, the dataset it belongs to, and its split. No further cleaning, tiling, relabelling or splitting is required before Milestone 3. In cases where a training run uses a smaller tile budget than the full prepared set, that sampling will be performed from the metadata (e.g. by dataset and building-pixel content) and recorded with the run, as planned in Milestone 1.

11. Challenges Encountered

The following lists the key challenges encountered and resolved:

- The Inria dataset in particular is a considerably large multi-gigabyte download of 5000x5000 GeoTIFFs. The inconvenience of the large file sizes was handled by downloading each dataset only once - centrally, to the team server (WHU 6.1 GB extracted, Inria 20 GB, Massachusetts about 1.3 GB). AIRS, which is much larger per tile (10000x10000), was not downloaded.
- There is a documentation inconsistency, where the WHU tile total differs between the official page text (8,189) and its own split lists (8,188). This discrepancy was recorded in Section 3, and ultimately resolved in favour of the split lists.

- There is licensing ambiguity with two of the four sources, i.e. Inria and Massachusetts, as they state no named license. This project handles the datasets with caution and explicitly for an academic use case, as declared in Section 4. The data is not redistributed.
- Challenges arose from format differences between GeoTIFF and PNG-style tiles, as well as encoding differences (e.g. 255 and 1 both representing a building). These issues were handled by the standardisation steps performed as part of preprocessing (Section 6).
- Due to buildings being a pixel minority, there is a major class imbalance. This imbalance was retained and documented per tile in the metadata, so that later sampling and metrics may account for it.
- The Massachusetts download was slowed by its source server, which limits parallel connections, so only part of its files had arrived when preprocessing first ran. The download was left running in the background and completed; the final prepared counts for Massachusetts are complete (604 tiles) and all final checks pass.
- On the team server, Python's multiprocessing pool (ProcessPoolExecutor) crashed under Python 3.14 (a forkserver transport error) during the first tiling run. The scripts were switched to a thread pool (ThreadPoolExecutor, 32 workers), after which all steps completed normally.
- One WHU training tile (tile 3508) was copied incompletely by an interrupted run, and the first metadata file was short by one row. The tile was re-copied from the raw download, its mask was untouched, and metadata.csv was regenerated; the final file has 23,372 rows, matching the tile count exactly.
- AIRS and WHU cover the same city, resulting in geographic overlap. However, the AIRS subset is test-only and is kept separate, and the overlap would also be reflected in the results.

However, some limitations still remain: the Inria split is same-city (in-dataset scores optimistic, Section 9); Massachusetts training labels carry the documented omission noise of roughly 5%; label definitions differ between datasets by design and are not harmonised. These are properties of the benchmark data, recorded here so later milestones interpret results against them.

12. Deliverables Produced

- Prepared datasets: WHU, Inria and Massachusetts in the common format of Section 7 (train/validation/test, images and masks as 512x512 PNG), stored on the team server in the prepared_dataset folder. AIRS is not part of the prepared dataset in this milestone.
- Metadata and data dictionary: metadata.csv with one row per prepared tile; its fields are defined in the schema table of Section 3, which serves as the data dictionary.
- Preprocessing scripts and checks: the step-by-step preparation scripts (inventory, pairing, splitting, mask inspection and standardisation, tiling, naming, metadata and EDA statistics, overlay generation and final verification) are kept with the project files, together with 9 split-list files (one per dataset and split) and 45 mask-overlay check images (5 per dataset and split).

13. Summary and Next Steps

In this milestone the datasets of the project were identified from their official sources, downloaded to the team server, described, and prepared into a single training-ready format: 512x512 PNG image-mask pairs with one folder structure, one mask encoding, one naming rule and one metadata index, giving 23,372 prepared tiles across the three core datasets. AIRS was not downloaded in this milestone; its small test-only subset is deferred to a later milestone.

Key observations from the data: the datasets differ exactly as the project needs them to - in resolution (0.3 m against 1 m), geography (New Zealand, the USA, Europe), and label definition (footprint-style labels in the core datasets; the roof-outline contrast dataset, AIRS, is deferred to a later milestone) - while their packaging differences (encodings, formats, split conventions) have been removed by preparation. Buildings are a pixel minority everywhere (11.00% to 18.66% of pixels per split), empty tiles reach 28.86% in WHU validation, Inria's labels require a documented substitute split, and the known label noise and overlaps are now written down where later analysis can see them.

The dataset is confirmed ready for model training: Milestone 3 (Model Architecture) can take the prepared folders and metadata as they are and set up the U-Net baseline and the end-to-end pipeline on top of them, with no further data preparation.

Contribution Log

Maintained for every milestone, as required in the Milestone 1 review. Each member confirms their own entry.

Team member	Contribution in this milestone
Bhawani Dutt	Dataset download Preprocess and Storage on the team server
Jaival	Exploratory Data Analysis
Prakhar Mishra	Documenting Preprocessing steps performed, Challenges Encountered and Deliverables.
Atharva Sarbhukan	Dataset Identification, Description, Governance for three datasets.
Abhishek Rawat	Data Integration, Data Augmentation

Sign-off

Each team member confirms that they have reviewed and approved this document.

Team member	Reviewed and approved	Date
Bhawani Dutt	Yes	29/10/2026
Jaival	Yes	29/10/2026
Prakhar Mishra	Yes	29/10/2026
Atharva Sarbhukan	Yes	29/10/2026
Abhishek Rawat	Yes	29/10/2026

References

Ji, S., Wei, S., & Lu, M. (2019). Fully Convolutional Networks for Multisource Building Extraction from an Open Aerial and Satellite Imagery Data Set. IEEE Transactions on Geoscience and Remote Sensing, 57(1), 574-586. DOI: 10.1109/TGRS.2018.2858817. Dataset page:https://gpcv.whu.edu.cn/data/building_dataset.html ;

deposit: https://doi.org/10.5281/zenodo.10043889

Maggiori, E., Tarabalka, Y., Charpiat, G., & Alliez, P. (2017). Can Semantic Labeling Methods Generalize to Any City? The Inria Aerial Image Labeling Benchmark. IEEE IGARSS, pp. 3226-3229. DOI: 10.1109/IGARSS.2017.8127684. Dataset: https://project.inria.fr/aerialimagelabeling/

Mnih, V. (2013). Machine Learning for Aerial Image Labeling. PhD thesis, University of Toronto. Dataset: https://www.cs.toronto.edu/~vmnih/data

Chen, Q., Wang, L., Wu, Y., Wu, G., Guo, Z., & Waslander, S. L. (2019). Aerial imagery for roof segmentation: A large-scale dataset towards automatic mapping of buildings. ISPRS Journal of Photogrammetry and Remote Sensing, 147, 42-55. DOI: 10.1016/j.isprsjprs.2018.11.011. Dataset: https://www.airsdataset.com/

WHU Building Dataset: Downloaded from the Zenodo repository:
https://zenodo.org/records/10043889
Inria Aerial Image Labeling Dataset: Obtained from the official Inria download page:
https://project.inria.fr/aerialimagelabeling/download/
Massachusetts Buildings Dataset: Downloaded from the University of Toronto dataset page:
https://www.cs.toronto.edu/~vmnih/data/
Massachusetts Building Shapefiles: Retrieved from:
https://www.cs.toronto.edu/~vmnih/data/mass_buildings/massachusetts_buildings_shape.zip
AIRS Dataset: Available at https://www.airsdataset.com/ (identified but not used in the final experiments).